Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/08/2023
PI Name:	Haveman, Natasha Ph.D.		
Project Title:	Developing a System for Rapid Diagnosis of Plant Diseases and Monitoring of Plant Microbiome for Spaceflight Applications		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: Dr. Natasha Haveman, formerly at the University of Florida (UF), joined the NASA Kennedy Space Center (KSC) Space Biology team in 2023. NOTE: name change to Haveman née Sng (Ed., June 2021)		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	01/01/2022	**End Date:**	12/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (University of Florida, Gainesville) Schuerger, Andrew Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC22K0160		
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Task Description:	Plants are integral to facilitating sustained human space exploration and colonization. Regardless of the scale, sustainable agriculture through bioregenerative life support systems or small plant-growth payloads needs to have proper mitigation strategies set in place to deal with potential microbial pathology issues in order to ensure crew safety on long-duration missions. However, to date, neither method nor tools currently exist to evaluate the plant microbial interactions or to diagnose potential plant diseases in spacecraft. The specific objective of this ground-based proposal is to use flight-compatible technologies to develop a method that will enable the rapid diagnosis of any plant diseases, as well as investigate changes in the microbiome of various plant diseases under a combination of space-relevant stressors. One major component of NASA's Space Biology mission directive is understanding plant-microbe interactions in the spaceflight environment. Several studies in the spaceflight environment and microgravity analogs have shown that these conditions can alter the plant-microbe interactions at both the genetic and physiological levels, often resulting in increased susceptibility to diseases. In addition, studies have shown that elevated carbon dioxide (CO2) and humidity, which are space-relevant stressors, can also increase the severity of plant diseases. The proposed plant microbiology research will address two focused areas defined by this NASA Research Announcement (NRA): (1) the combined effects of space-relevant stressors and (2) the commonality and differences across plant species. Our central hypothesis is that the combined use of WetLab2/µTitan, VloTRAX-V2, and MinION will enable a streamlined, semi-autonomous, and spaceflight-compatible method to identify the impact of space-relevant stressors (i.e., elevated CO2 and high humidity) on plant microbiomes and rapidly diagnose plant diseases for on-orbit applications. The aims of this proposal are: (1) Optimize the workflow for WetLab2 nucleic acid extraction protocol, library preparation with VloTRAX-V2, and MinION sequencing in a variety of plant species; and (2) Compare and contrast infection symptoms of four pathosystems grown under space-relevant stressors versus normal growth conditions and use the optimized workflow to diagnose various phytopathogens and monitor changes in the microbial population under those conditions. The pathosystems proposed herein cover the three main categories of phytopathogens (i.e., fungi, bacteria, and viruses) that cause common plant diseases such as wilts, chlorosis, and soft rots. These pathosystems are Mizuna - Erwinia carotovora (soft rot); Tomato - Fusarium oxysporum (wilt); lettuce - Pythium aphanidermatum (root rot); and Chimayo Chilli - tomato mosaic virus (chlorosis). Commonalities and differences in the optimized protocol for each of the pathosystems will be recorded so that the pipeline can be designed to effectively diagnose a variety of plant diseases relevant to NASA's agricultural objectives. The expected results from this proposal will: (1) Improve the use of currently available flight-compatible technologies for on-orbit assessment of plant-microbial interaction and (2) Tease apart the combined effects of elevated CO2 and humidity levels on the plant microbiome and pathogenesis in the International Space Station (ISS) environment. This study will not only provide a pipeline to diagnose plant diseases and monitor plant microbiomes using spaceflight-compatible tools, but it will also provide detailed procedures for harvesting different plant tissues for various common diseases along with calculations for the mass consumption for each diagnosis. Since the MinION sequencer will be used on station for future sequencing experiments (i.e., Microbial Tracking-3) new methods should be developed to allow for MinION's versatility to be exploited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although this project intended to develop a quick and easy method of extracting plant pathogen nucleic acid for spaceflight applications, there are many applications on Earth that would benefit from this methodology. Onsite plant disease diagnosis among field specialists, crop consultants, and growers can help prevent the further spread and substantial losses in the multibillion-dollar agricultural industry each year. Further maturity of this project will allow for on-site point-of-care crop disease diagnostics at farms or research and educational centers.
Task Progress:	Significance The rationale behind this proposal is to advance the technical readiness level (TRL) necessary to support a sustainable agriculture system for future human space exploration. The goal will be accomplished by developing tools to establish a strong foundation in understanding how plant-microbe interactions impact the safe and successful production of plants in orbit. From this project, one manuscript is in preparation for peer-reviewed publication, and one oral presentation was given at the American Society for Gravitational and Space Research (ASGSR) annual meeting (2022). --Manuscript: Haveman N.J., Schuerger A.C., Yu P-L., Brown M., Doebler R., Paul A-L., Ferl R.J. Advancing the Automation of Plant Nucleic Acid Extraction for Rapid Diagnosis of Plant Diseases in Space. Frontiers in Plant Science (In preparation) --Oral Presentation: 2022 American Society for Gravitational and Space Research (ASGSR), Houston, TX. Haveman NJ, Schuerger AC, Yu P-L, Paul A-L, Ferl RJ. Automation of a plant nucleic acid extraction system for on-orbit plant disease diagnosis. (Technical session: Plants) Task Progress The major outcome of this one-year project is the establishment of a method that enables the extraction of plant and plant pathogen nucleic acids that is well suited for use with the MinION sequencing platform for the diagnosis of plant diseases. Although the initial intent was to collaborate with the WetLab-2 implementation team to develop the protocol, the program was decommissioned before the start of this proposal. We proceeded to collaborate directly with the company, Claremont BioSolutions LLC (CBIO), which aided WetLab-2 with their protocols. The process of optimizing a protocol with potential compatibility with CBIO's SimplePrep® automated platform required testing various compatible devices, lysis buffers, tissue sizes, incubation times, and downstream procedures. Ultimately, using the microHomogenizerTM device, with TES lysis buffer [0.2 M Tris-HC1 [pH 8], 10 mM EDTA [pH 8], 0.5 M NaC1, 1% SDS], tissue sizes that were approximately 2-3 mm, with at least five minutes of homogenization time, and precipitation of DNA with 500 µl of isopropanol yielded the best results. To meet the goals of Objective 1– during testing, DNA concentrations were measured with QubitTM dsDNA broad range (BR) kit and used as an indicator of success. To meet the goals of Objective 2 – three different pathosystems, plant hosts and phytopathogens, were used to test the feasibility of using the developed method to extract nucleic acids for diagnosis of plant diseases. Here, Solanum lycopersicum cv. 'Red Robin' (Tomato), Lactuca sativa cv. 'Outredgeous Red Romaine' (Lettuce), and Capsicum annuum cv. 'Chablis' (Pepper) plants were inoculated with a plant fungal pathogen Fusarium oxysporum f. sp. lycopersici (FOL), an oomycete pathogen Pythium aphanidermatum P1771, and a plant viral pathogen Tomato Mosaic Virus (ToMV), respectively. Polymerase chain reaction (PCR) amplifications with specific primers were performed to validate the presence of the

	various pathogens from each of the tissue samples listed above. Results showed that the developed protocol was able to isolate pathogenic nucleic acids that could be amplified through PCR. Downstream nanopore sequencing showed the feasibility of this method to diagnose plant diseases in future spaceflight applications. DNA extracted from various tissue types (leaves, stem, and roots) of pathosystem #2 (Tomato-FOL), were sequenced with the MinION nanopore sequencer to see if the results obtained corroborated with those from the PCR reactions. The quality score of the reads obtained showed successful sequencing with an average mean read length for libraries at approximately 2688 bp and the mean Q-scores were > 10. Taxonomy classification Kraken2 reports were sorted based on the highest percentage of fragments covered by the clade and the rank code (S) for species. The top 12 species with the highest percentage of = 0.1% were selected from each sample. Results showed that only inoculated tomato tissue showed the presence of FOL. Thus, the work presented here demonstrates a methodology that would lay the foundation for future plant disease detection in spaceflight modules. Prompt onsite disease diagnosis is an important aspect of ensuring food safety and security for future human space missions.
Bibliography Type:	Description: (Last Updated: 08/03/2023)
Abstracts for Journals and Proceedings	Haveman NJ, Schuerger AC, Yu P-L, Paul A-L, Ferl RJ. "Automation of a plant nucleic acid extraction system for on-orbit plant disease diagnosis." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Articles in Peer-reviewed Journals	Haveman NJ, Schuerger AC, Yu P-L, Brown M, Doebler R, Paul A-L, Ferl RJ. "Advancing the automation of plant nucleic acid extraction for rapid diagnosis of plant diseases in space." Front Plant Sci. 2023 Jun 14;14:1194753. https://doi.org/10.3389/fpls.2023.1194753 ; PMID: 37389293; PMCID: PMC10304293 , Jun-2023